CAP V-SAR | Civil Air Patrol Virtual Search and Rescue Trainer

AI Mission Pilot
Concept of Operations

Mission Observer practice with a virtual Mission Pilot

AI Mission Pilot Concept of Operations | Draft 0.1 | 10 September 2026

This document defines how an AI Mission Pilot can support Mission Observer practice in CAP V-SAR while preserving pilot authority, trainee decision making, hidden scenario truth and instructor judgment. The recommended operating model is a virtual crewmate built from a conversational layer, a strict maneuver gate and one replaceable flight-control backend.

1 Executive intent

CAP V-SAR should add an AI Mission Pilot so a Mission Observer can practise a complete electronic-search sortie when a qualified human Mission Pilot is unavailable. The AI MP performs the pilot-side work needed to make the exercise coherent: it maintains the simulated aircraft, receives and reads back tactical directions, manages its assigned radio work and reports whether a maneuver was actually achieved.

The AI MP is a training aid inside a simulation. It does not confer qualification, replace instructor or Skills Evaluation Trainer judgment, create operational authority, or represent an approved Civil Air Patrol program. Its purpose is repeated deliberate practice and evidence-rich debriefing.

The governing design decision is separation of responsibilities. A language model may interpret conversation, but it never writes aircraft controls directly. A deterministic gate validates each maneuver against authority, sequence, units, current aircraft state and the exercise envelope. A flight backend executes the accepted maneuver and reports measured progress. The instructor alone retains scenario truth and adjudicates training performance.

2 Operational need

Electronic search is a crew task. The Mission Observer must manage navigation, search geometry, the Becker SAR-DF 517 or RHOTHETA RT-600, the VHF-FM mission net, observations and reports while coordinating continuously with the pilot. Simulator practice is constrained when an experienced pilot must be present for every event, even when the lesson is centered on Observer decisions rather than stick-and-rudder skill.

The desired outcome is on-demand MO practice that still feels like a crewed sortie. The trainee directs the mission and works the search problem. The AI MP carries the pilot workload, challenges unsafe or ambiguous requests, and communicates aircraft state without solving the search. The instructor controls the scenario and debriefs the recorded decisions.

3 Roles and authority

Role	Operational responsibility	Authority boundary
Mission Observer trainee	Plans and directs the search; operates DF and mission radio; interprets signals; logs and reports results.	May request maneuvers within the briefed exercise block. Does not override flight-protection refusals.
AI Mission Pilot	Emulates pilot-side crew coordination, flies accepted maneuvers, performs readbacks and reports achieved or unable.	Uses only trainee-visible information and aircraft observations. Does not self-clear, reveal target truth or award credit.
Instructor or SET	Briefs the task, configures effects, observes performance, reveals truth in debrief and makes training judgments.	May pause, terminate, reposition or take direct control at any time.
Human safety operator	Monitors the simulator and can disconnect automation or take controls.	Has the highest control priority. Takeover revokes all AI command authority.
ATC actor	Provides clearances, restrictions and traffic sequencing in the chosen ATC mode.	Only this actor issues clearances. The AI MP may request, read back and comply.

In a real aircraft the Mission Pilot remains pilot in command. In CAP V-SAR the AI MP emulates that crew function, while the human instructor or safety operator owns the simulator session and can intervene without negotiation.

4 How a training sortie operates

Phase	Participants	What happens
Brief	Instructor, trainee, AI MP	The instructor issues a trainee-visible tasking. The trainee plans the search and briefs roles, route, altitude block, communications and abort conditions.
Setup	Trainee, safety operator	CAP 487 is positioned and configured. Radios and DF instruments are checked. Other flight-control writers are stopped. The AI MP acquires one control lease.
Execution	Trainee, AI MP, ATC or base as configured	The trainee gives explicit directions. The AI MP reads back, validates, executes and reports measured capture. The instructor injects effects without coaching unless the selected mode permits it.
Takeover or fault	Safety operator, AI MP	A human takeover revokes the lease before requesting autopilot disconnect. A telemetry or envelope fault revokes the lease and requests the configured simulator-safe response.
Debrief	Instructor or SET, trainee	Beacon truth is revealed. Requests, interpretations, readbacks, actual track, DF state, assistance and interventions are reviewed together.

4.1 The command cycle

Every flight request follows one visible sequence: heard, interpreted, validated, read back, accepted, executing and achieved. A request may instead be refused with a reason. The system does not say “established” merely because it wrote a heading bug or altitude selector; it waits for measured aircraft performance to settle within the configured tolerance.

- “Heading zero niner zero magnetic” becomes a magnetic-heading target. A bare “zero four five” is ambiguous and is not executed.
- “Do not turn left” and “heading zero three zero, not one three zero” require clarification. Negation is never discarded.
- A directional command such as “turn left heading zero niner zero” is accepted only when the active backend supports prescribed turn direction.
- A correction supersedes the active command through a new sequence number. Old, duplicated or delayed messages cannot replay.

5 Training modes

The instructor selects the amount of assistance before the sortie. The mode is recorded with the evidence so a coached repetition is never mistaken for an evaluation attempt.

Mode	AI MP behavior	Use
Coaching	Explains refusals, provides hints when asked, and may prompt an overdue procedural action. Every hint is recorded.	Initial equipment and crew-coordination practice.
Practice	Responds to directions and normal pilot duties. It avoids unsolicited tactical advice while retaining flight protection.	Repetition after instruction and scenario variations.
Evaluation	Provides only operational readbacks, aircraft reports and required safety interventions. It does not hint at the search solution.	Instructor-observed demonstration against an approved rubric.

5.1 Automation follows the lesson

Automation must match the skill being practised. When the objective is vectoring and turn anticipation, the AI MP flies only the headings and altitudes the MO requests. When the objective is search management, it may track an MO-approved pattern from the flight planner while the trainee monitors coverage and retasks. Automatic beacon homing is reserved for demonstrations because it would otherwise solve the trainee's problem.

The same principle applies to radios. VHF-FM operation remains the MO's responsibility when radio management is part of the lesson. The AI MP handles pilot-side ATC communications only in the configured ATC mode.

6 Information boundaries

The AI MP receives an explicit pilot-facing briefing rather than raw scenario data. The allowlist contains only what a human pilot would be briefed: exercise identifier, aircraft identity, approved route or search area, altitude block, communications plan, weather assumptions, operating mode and termination criteria.

Available to AI MP	Kept from AI MP
CAP 487 aircraft observations and selected cockpit indications	Beacon latitude, longitude and elevation
Trainee-visible tasking and approved route or pattern labels	Instructor notes, answer keys and unrevealed scenario effects
Assigned altitude block, airspace limits and communications plan	Propagation debug values and instructor-only distance or bearing truth
Actual DF indications when the pilot role would reasonably observe them	A separately computed perfect bearing or homing solution

The evaluator may receive hidden truth after or during the exercise as required for scoring, but it is a separate process and prompt. Changing hidden beacon truth while keeping all pilot-visible observations the same must not change the AI MP's tactical response.

7 System architecture

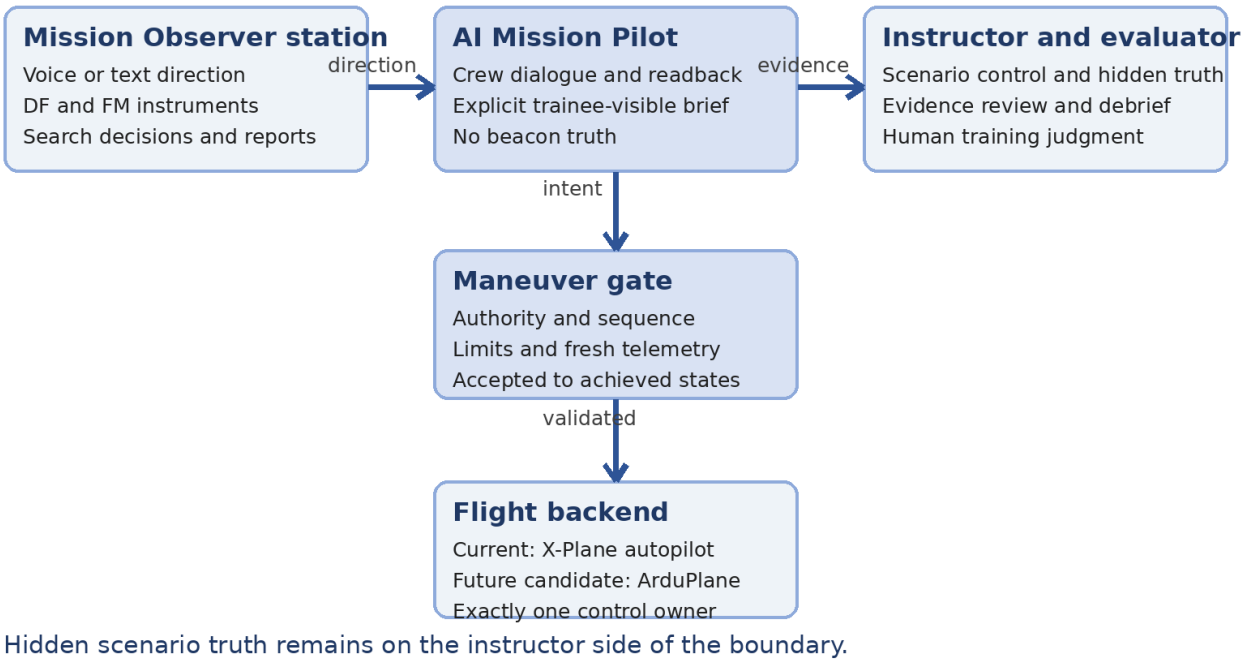


Figure 1 AI Mission Pilot information and control boundary

7.1 Layered controller model

The controller follows the same broad separation used in mature fixed-wing autopilot systems. Mission direction produces a navigation target; guidance turns the target into an aircraft command; the aircraft autopilot stabilizes the maneuver; monitoring closes the loop with measured state. Conversation remains above that stack.

Layer	Responsibility	Initial implementation
Crew dialogue	Interpret speech or text, preserve qualifiers, ask for clarification and deliver a concise readback.	Text first; local speech services later.
Maneuver gate	Own authority, sequence, units, limits, telemetry freshness and command state.	Implemented in services/ai_pilot/maneuvers.py.
Guidance backend	Select heading and indicated altitude, release controls and report observations.	X-Plane autopilot adapter.
Aircraft stabilization	Continuously hold the selected lateral and vertical modes.	CAP 487 simulated autopilot in X-Plane 12.
Evidence and evaluation	Combine dialogue, command state, aircraft track and instrument evidence for debrief.	Planned integration with the exercise console.

7.2 ArduPlane compatibility

ArduPlane is a candidate replacement backend, not a dependency. A bounded trial must confirm X-Plane 12 compatibility, CAP 487 tuning, cockpit indications and safe control transfer. The shared backend contract permits that trial without changing MO-facing commands. Only one backend may command the aircraft; the simulated Garmin autopilot and ArduPlane must never drive the same control surfaces at the same time.

8 Command and control rules

Rule	Operational meaning
Explicit reference	Headings are magnetic unless another supported reference is explicitly named. Altitudes specify indicated MSL or AGL; unsupported references are refused.
One maneuver at a time	Compound directions are split and read back separately unless the backend explicitly supports a coordinated maneuver.
No silent normalization	A heading outside 000 through 360, an impossible altitude or an unknown unit is refused rather than wrapped, clamped or guessed.
Measured completion	Accepted, executing and achieved are distinct. Achievement requires stable observed capture.
Single control owner	A lease and monotonic sequence protect against duplicate writers, replay and stale commands.
Human priority	Takeover revokes the lease first, then requests autopilot disconnect. The human confirms the cockpit state.

8.1 Standard readbacks

- Accepted: “Selected heading zero niner zero magnetic, altitude four thousand five hundred feet indicated. Awaiting measured capture.”
- Achieved: “Established heading zero niner zero, four thousand five hundred.”
- Ambiguous: “Confirm magnetic heading zero four five.”
- Refused: “Unable descent. Requested altitude is outside the assigned block.”
- Takeover: “Control revoked. Autopilot disconnect requested; verify cockpit indications.”

Fast, templated readbacks should be generated from the accepted structured command. A language model may handle ordinary crew conversation, but it does not invent the authoritative readback or alter a command after validation.

9 Communications and ATC

Intercom, aeronautical radio and VHF-FM traffic are separate audio and authority paths. Intercom speech is clean. Radio audio uses the selected receiver, isolation, split-COM and microphone routing already represented in the CAP V-SAR dataref contract. The AI MP must not recognize its own synthesized speech or received radio traffic as a cockpit command.

ATC mode	Behavior
Scripted offline	A scenario service supplies known clearances and traffic events. The AI MP requests and reads back them through a deterministic state machine.
Human facilitator	An instructor or remote participant acts as ATC. The AI MP treats that authenticated audio source as the clearance authority.
Live human ATC	PilotEdge or another approved exercise arrangement provides real controllers. The AI MP transmits and complies within the exercise brief.

The AI MP never says “cleared for takeoff” on its own initiative. It may report that a clearance was received only after the configured ATC actor supplies it. Mission radio names and frequencies come from the exercise communications plan and CAP channel card; they are not embedded in dialogue prompts.

10 Safety and failure handling

Safety in this CONOPS means controlled simulator behavior and protection of training integrity. It does not imply aircraft certification or operational airworthiness. A deterministic loop can still act incorrectly if its sensors are stale, its control ownership is unclear or its flight limits are incomplete.

Condition	Required response	Evidence
Human takeover	Revoke AI authority, discard queued commands and request autopilot disconnect. Human verifies the flight director and servos.	Takeover time, prior command, disconnect request and subsequent aircraft state.
Stale or missing telemetry	Stop accepting commands, revoke authority and request the configured simulator pause or validated holding response.	Last sample ages, affected fields and response outcome.
Envelope excursion	Refuse new maneuvering, announce the reason and execute only a prevalidated protection action.	Airspeed, AGL, bank, vertical rate and active limits.
Speech or model outage	Continue the already accepted deterministic maneuver or transition to the prebriefed fallback. Offer text controls if available.	Service state, current command and fallback selected.
Conflicting controller	Prevent acquisition when detectable. Stop the session if a second writer cannot be excluded.	Owner identity, lease state and detected writes.
Network loss	Keep local flight protection independent of remote model and exercise services; do not infer search cues from missing data.	Connection state, local controller state and any pause.

10.1 Exercise limits

Limits are selected for each exercise and aircraft configuration. The current prototype uses a configurable altitude block plus provisional airborne checks for indicated airspeed, AGL and bank. These values are not POH limitations, regulatory minimums or terrain-clearance assurance. Before operational training, protection must include projected-path terrain lookahead, airspace constraints, phase awareness, altimeter setting and a validated escape or pause policy.

11 Training evidence and evaluation

The system records evidence for human review rather than producing an automatic qualification. CAPF 91 is a Mission Pilot flight check and is not the MO scorecard. An experienced instructor or SET must map the evidence to the applicable Mission Observer tasks and decide what, if any, credit is appropriate.

Evidence stream	What it supports
Audio and transcript	What the trainee actually said, what speech recognition produced and whether a correction or ambiguity was handled.
Structured command log	Parsed intent, reference, units, source, sequence, validation, readback and command-state transitions.
Aircraft track and modes	Whether the requested maneuver was flown, when it was achieved and whether the pilot system introduced an error.
DF and radio state	Receiver setup, signal availability, squelch, mode, channel, PTT route and the indications available to the trainee.
Assistance and interventions	Every hint, refusal, safety action, pause and human takeover, separated from trainee performance.
Scenario truth	Used by the instructor during debrief to compare decisions and track with the actual beacon and injected effects.

11.1 Measures that matter

Useful measures include timely and unambiguous pilot direction, correct response to signal loss, receiver setup discipline, recognition of reciprocal or multipath behavior, adherence to assigned altitude and airspace constraints, communications coordination, coverage monitoring and recovery from a wrong hypothesis. System errors are scored separately from trainee errors. Simulated time, wall-clock time, pauses, scenario identifier and random seed travel with every record.

12 Current implementation status

As of 10 September 2026, CAP V-SAR includes a separate airborne maneuver deck in services/ai_pilot. It provides a backend interface, X-Plane autopilot adapter, strict text grammar, control lease, increasing command sequence, configurable exercise envelope, telemetry-age checks, measured capture and a local takeover control. Automated gate, backend-intent and HTTP tests pass.

Status	Capability
Current	Text commands for magnetic heading and indicated altitude through the isolated maneuver deck on loopback port 8089.
Current but unverified live	X-Plane autopilot selection, telemetry monitoring, pause request and disconnect request. Simulator handling and cockpit indications have not yet passed a live acceptance flight.
Legacy only	Scripted KSEE-to-KRNM automation, direct chat keyword actions and direct-write web or LLM tool routes. These bypass the new gate and must remain stopped during maneuver-deck use.
Planned	Voice intercom, scenario briefing allowlist, planner route tracking, radio and ATC state, evaluator integration, terrain lookahead and a future ArduPlane feasibility trial.
Outside initial release	Autonomous takeoff and landing, contour or canyon following, unrestricted natural-language control and automatic beacon homing during evaluation.

13 Delivery plan and acceptance gates

Gate	Scope	Exit evidence
A	Supervised live maneuver trial	CAP 487 maintains commanded headings and indicated altitudes over representative winds and update rates; takeover, stale telemetry and pause behavior are observed in X-Plane.
B	MO vectoring session	A trainee completes an airborne electronic-search problem using text direction only. Hidden-truth isolation and the complete evidence record are demonstrated.
C	Planner route tracking	The AI MP tracks labeled planner legs while preserving MO authority to approve, amend and terminate the pattern.
D	Local voice and cockpit audio	Speech latency and command accuracy meet measured median and 95th-percentile targets on the actual headset and hardware. Radio traffic cannot trigger commands.

Gate	Scope	Exit evidence
E	Communications and ATC	Intercom, COM and VHF-FM routing match cockpit selections. ATC clearances come only from the configured actor.
F	Evaluation-mode pilot	Qualified instructors confirm that the AI MP supports the lesson without coaching, disclosing truth or masking trainee errors.
G	ArduPlane decision	A bounded SITL trial either demonstrates X-Plane 12 compatibility, CAP 487 tuning and coherent takeover, or records why the X-Plane autopilot remains the backend.

13.1 Minimum acceptance cases

- Negation and correction: “Do not take off,” “cancel that,” and “zero three zero, not one three zero” produce no unintended aircraft write.
- Replay resistance: duplicate, delayed and pre-takeover commands cannot re-engage control.
- Partial observability: changing hidden beacon truth without changing observable instruments does not change the pilot’s response.
- Communications isolation: a deselected microphone cannot transmit, and received traffic or synthesized speech cannot become a flight instruction.
- Service failure: loss of speech, model or exercise server preserves the prebriefed deterministic flight response and leaves no orphaned controller.
- Training validity: experienced Mission Pilots and Observer instructors judge the virtual crewmate credible enough for the intended MO lesson before trainee rollout.

14 Representative use in CAP V-SAR

Scenario C 1 near McClellan Palomar is the capstone use case. A 121.5 MHz signal is reported in the approach area without a known position. SoCal Approach protects the published approach altitude and assigns the search aircraft blocks above and below arriving traffic. The MO must plan, negotiate, direct and revise the search while interpreting signal behavior and maintaining a useful geometry.

In coaching mode, the AI MP may explain an ambiguous direction or remind the trainee which altitude block is active. In practice mode it reads back and flies accepted directions but does not propose the search solution. In evaluation mode it performs normal pilot coordination and intervenes only for the configured flight-protection conditions. It never receives the beacon coordinates.

A representative exchange is intentionally ordinary. The MO says, “Pilot, heading zero three zero magnetic, maintain four thousand five hundred feet indicated.” The AI MP validates the assigned block, selects the maneuver and reads it back. It later reports established only when the measured aircraft state is stable. If ATC changes the block, that authenticated clearance updates the permitted envelope; the AI MP does not invent the change.

The debrief overlays the actual track, trainee directions, DF indications, ATC restrictions, pilot-system responses and hidden beacon truth. The instructor can then separate an Observer decision error from an AI pilot execution error and rerun the same geometry with a different beacon or propagation effect.

15 Decisions required before trainee use

Decision	Owner
Approve the intended training tasks and rubric for coaching, practice and evaluation modes.	Instructor lead and SET
Approve CAP 487 airborne exercise limits, terrain-lookahead policy and the response to telemetry loss.	Mission Pilot SMEs and safety lead
Choose the first ATC mode and identify the authenticated clearance source.	Exercise director and communications lead
Decide whether the X-Plane autopilot is sufficient or authorize the bounded ArduPlane feasibility trial.	Technical lead with Mission Pilot validators
Define retention, access and review rules for audio, transcript and evaluation evidence.	Project officer and training leadership

16 References and terms

Project references

- CAP V-SAR Concept of Operations, docs/ragd/vaultnotes/CONOPS-cap-vsar.md.
- CAP V-SAR dataref contract, docs/DATAREF_CONTRACT.md.
- AI Pilot service status and operating limits, services/ai_pilot/README.md.
- CAP V-SAR Implementation Plan version 0.8 and Concept and Implementation Summary.

External technical references

- ArduPilot Plane Architecture Overview
- ArduPilot Using SITL with X-Plane 10/11
- Laminar Research: Accessing the X-Plane Autopilot from Datarefs
- Civil Air Patrol Mission Pilot Qualifications and Endorsements

Term	Meaning in this document
AI MP	A virtual crewmate that emulates selected Mission Pilot duties inside CAP V-SAR.
MO	Mission Observer trainee directing the electronic-search task.
SET	Skills Evaluation Trainer responsible for human evaluation and any task signoff.
Backend	The single component permitted to translate validated maneuvers into simulator flight-control actions.
Hidden truth	Scenario information unavailable to the trainee, including actual beacon location and instructor-only effects.
Control lease	Revocable authority token that allows one command source to submit an ordered maneuver sequence.

Document status: Draft concept for CAP V-SAR project review. Not an approved Civil Air Patrol program, qualification device or substitute for aircraft training.